

Single Subject

- Started with single subject
 - [Dataset](#)
 - Started with '001'
- Using MNE
- Initial steps
 - Load data
 - Notch filter at 60 Hz and highpass at 1 Hz
 - Manually selection of bad channels
 - Channels are noisy but maybe still useful, will try again on later iterations
 - Tried with both manually and automatic (mne_icalabel) ICA component exclusion
 - Bandpass between 8 Hz and 30 Hz
 - Epoch from 0 s to 1.5 s with -2 s to 0 s removed as baseline
 - Split epochs by trial type
 - No epoch removal
 - Looked at left/right channel alpha power

Multi-Subject

- Moved on to using multiple subjects
- Parameters:
 - `subjects = ['001','002','003']`
 - `bads = {'001':['Iz','Oz','O1','O2','Poz','Po3','Po4','Po7','Po8'], '002':['Iz','Oz','O1','O2','Poz','Po3','Po4','Po7','Po8','Af7','T7'], '003':['T7','T8']}`
 - `components = {'001':15, '002':20, '003':15}`
 - `ica_excludes = {'001':[0,1], '002':[0,1], '003':[0,1,3,5,6,7,13,14]}`
- Attempted manual channel and ICA component selection with subjects 1, 2, and 3

Multi-Subject pt. 2

- Reverted to automatic ICA component labeling and no removal of channels
- Ran for all 109 subject for task 3 (open and close left/right hand), subject 92 failed for some numpy reshaping issue
- Steps:

- Load data
- Notch filter at 60 Hz and highpass at 1 Hz
- All channels kept
- Run ICA, label components with `mne_icalabel` and remove all non-brain components
- Run Artifact Subspace Reconstruction (ASR) to attempt to remove any remaining large artifacts, somewhat successful
- At first was running bandpass on data from 8 Hz to 30 Hz but strongly considering removal due to possible overlap with below steps
- Compute PSD for each epoch and take the power values from 8 to 13 Hz
- Average power over band, and over selected channels
 - `left_channels = ["Fc1", "Fc3", "C1", "C3", "Cp1", "Cp3"]`
 - `right_channels = ["Fc2", "Fc4", "C2", "C4", "Cp2", "Cp4"]`
- Examined left brain and right brain power in left hand and right hand epochs
- Plotted left/right brain power difference for both types of trials for all subjects